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In [1]: import pandas as pd
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In [2]: # Sample data
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [10000, 12000, 15000, 13000, 17000, 16000],
    "Profit": [2000, 3000, 4000, 2500, 3500, 3000]
}
```

```
In [3]: df = pd.DataFrame(data)
```

```
In [4]: df
```

```
Out[4]:
```

	Month	Sales	Profit
0	Jan	10000	2000
1	Feb	12000	3000
2	Mar	15000	4000
3	Apr	13000	2500
4	May	17000	3500
5	Jun	16000	3000

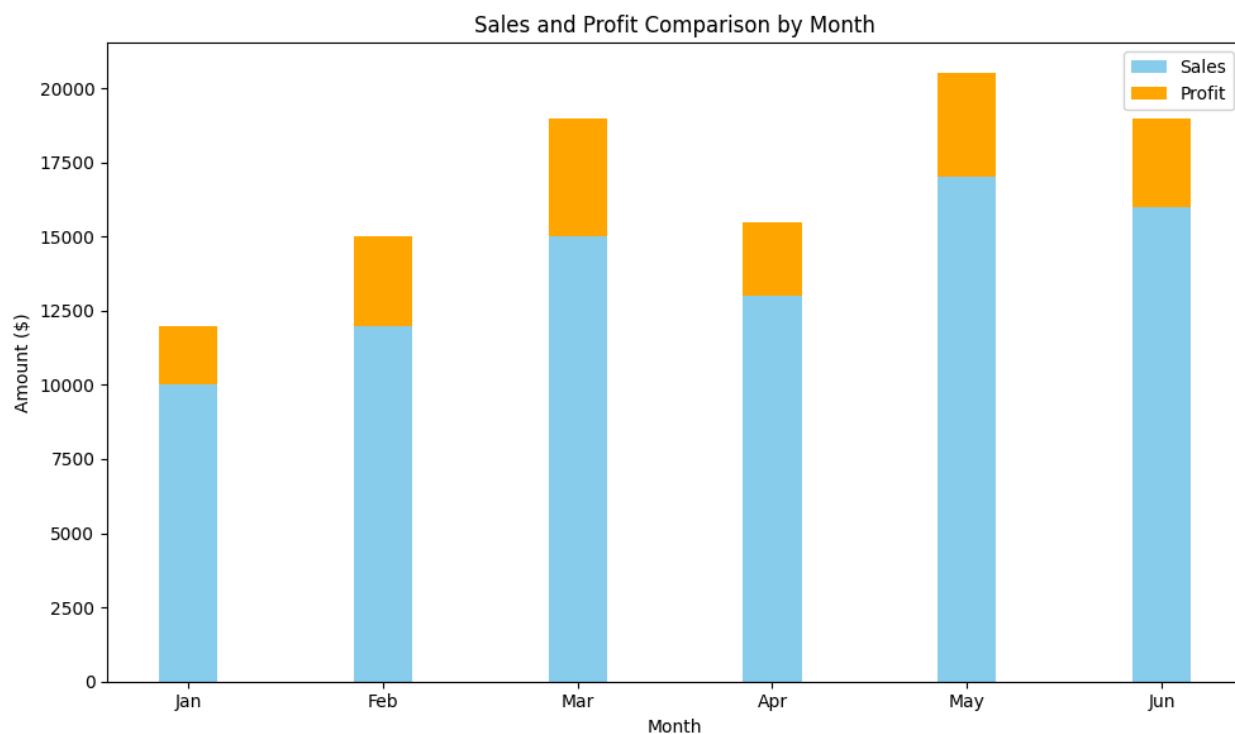
```
In [6]: # 1- line plot sales over time

import matplotlib.pyplot as plt

# Line Plot: Sales over Time
plt.figure(figsize=(8, 5))
plt.plot(df['Month'], df['Sales'], color='blue', marker='o', linestyle='-', label='Sales')
plt.title('Sales Trend Over Months')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

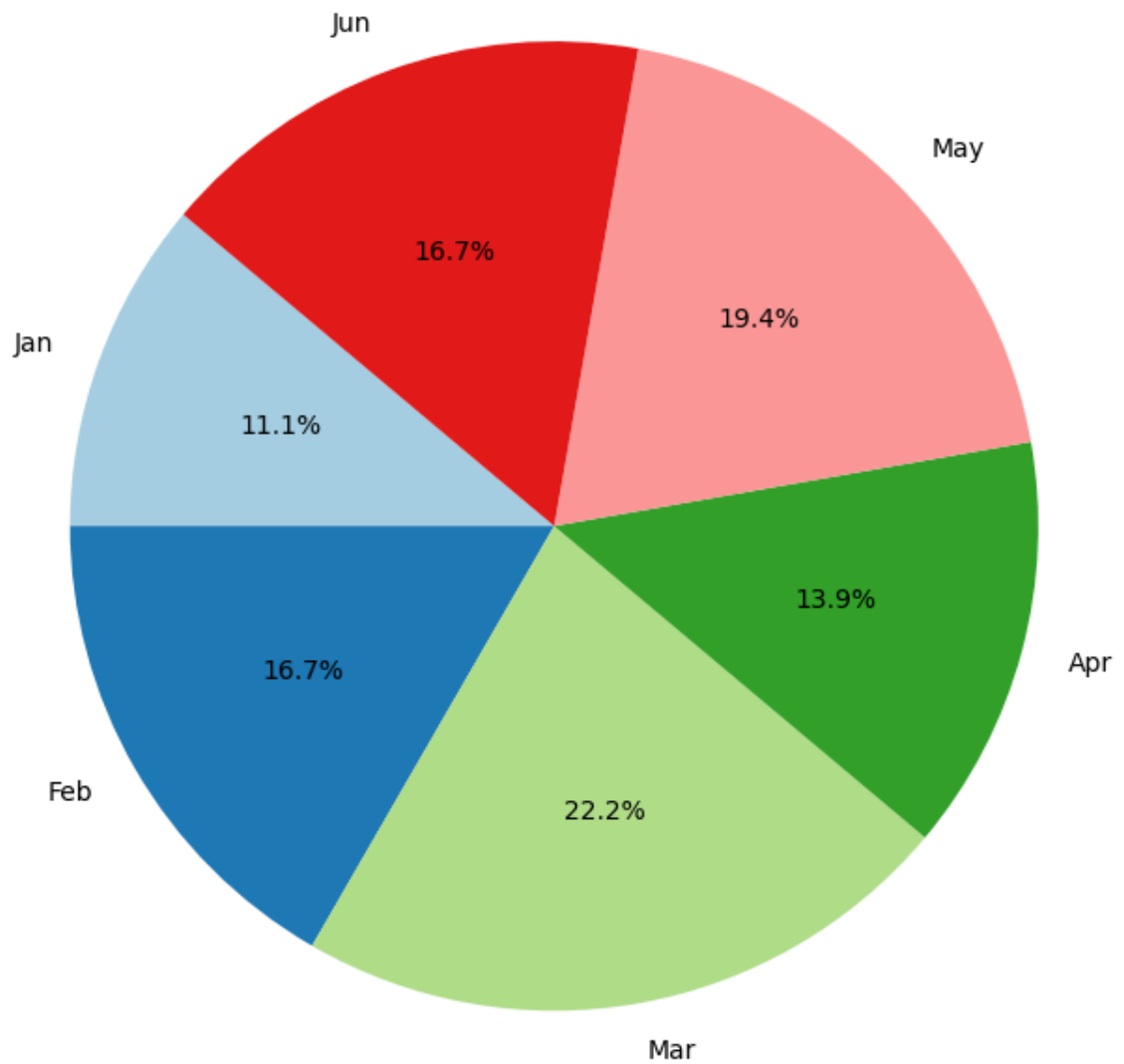


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In [7]: # Bar Plot: Sales vs Profit by Month
plt.figure(figsize=(10, 6))
width = 0.3
plt.bar(df['Month'], df['Sales'], width=width, label='Sales', color='skyblue')
plt.bar(df['Month'], df['Profit'], width=width, label='Profit', color='orange')
plt.title('Sales and Profit Comparison by Month')
plt.xlabel('Month')
plt.ylabel('Amount ($)')
plt.legend()
plt.tight_layout()
plt.show()
```

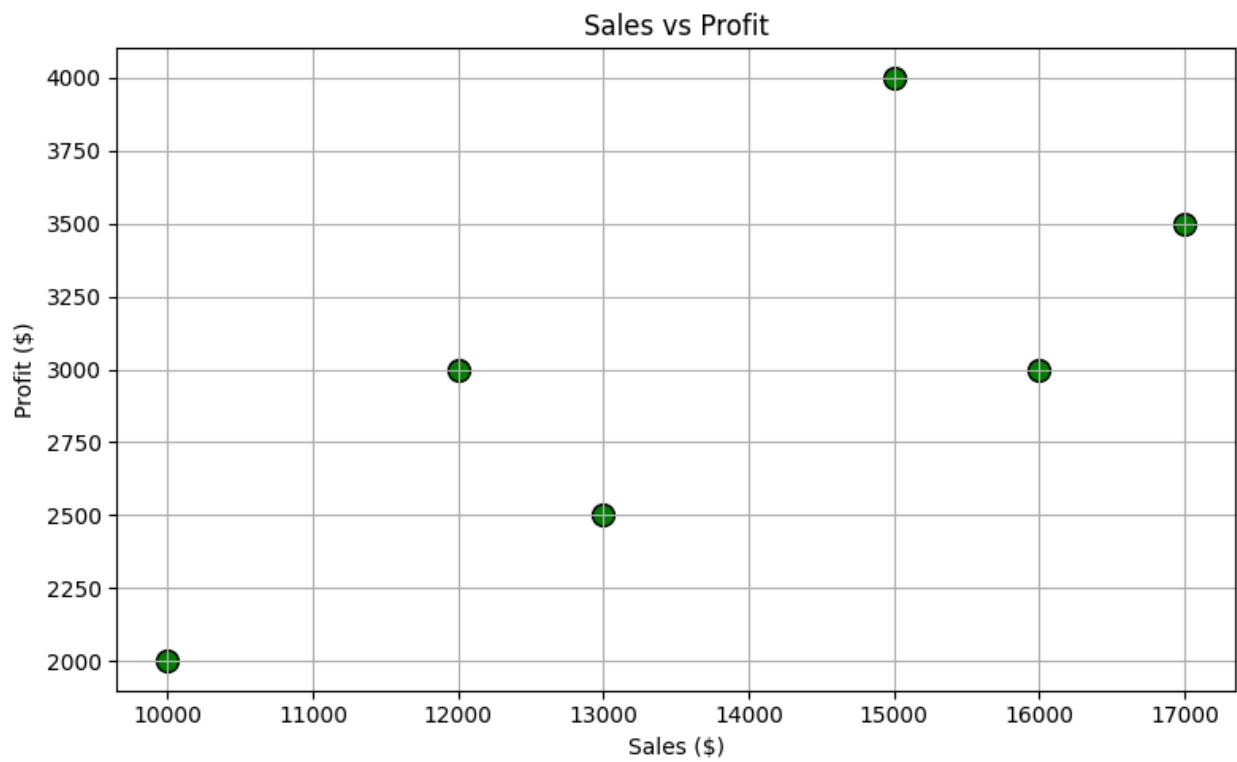


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In [8]: # Pie Chart: Profit Distribution by Month
plt.figure(figsize=(7, 7))
plt.pie(df['Profit'], labels=df['Month'], autopct='%1.1f%%', startangle=140, c
plt.title('Profit Distribution by Month')
plt.tight_layout()
plt.show()
```

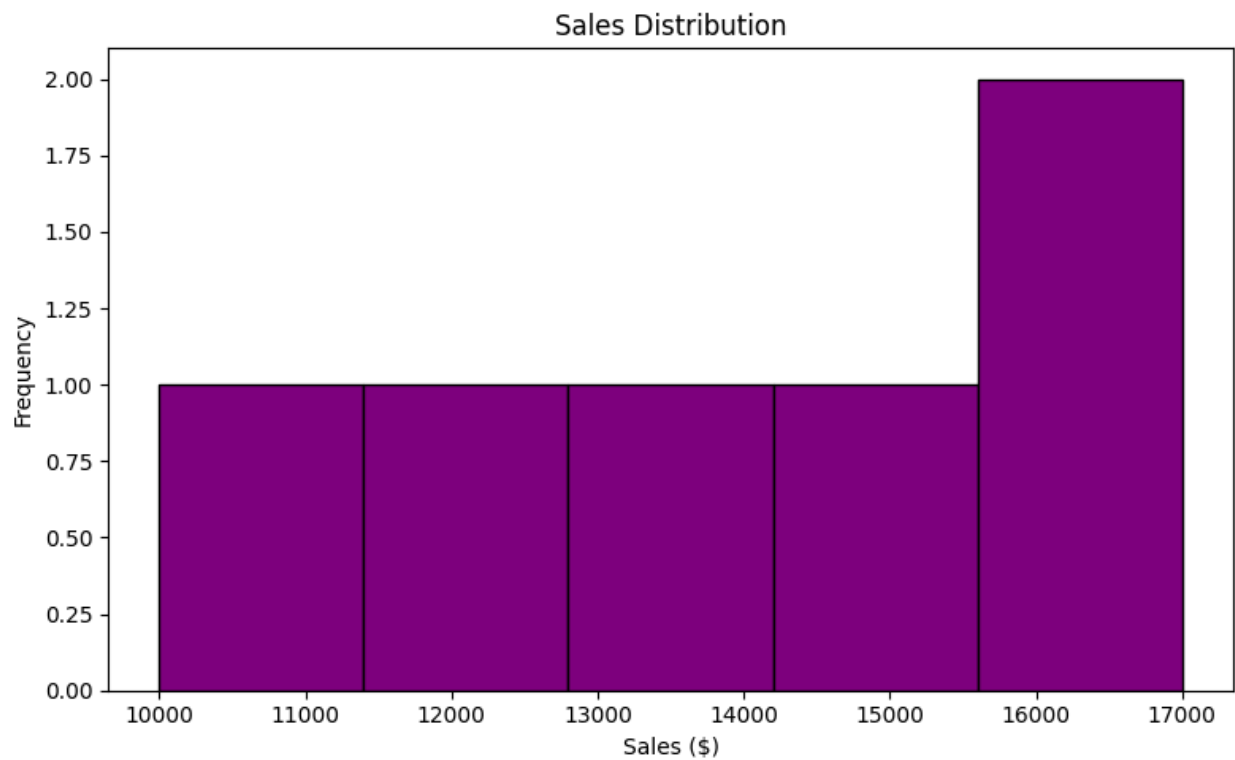
Profit Distribution by Month



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In [9]: # Scatter Plot: Sales vs Profit (Correlation)
plt.figure(figsize=(8, 5))
plt.scatter(df['Sales'], df['Profit'], color='green', s=100, edgecolors='black')
plt.title('Sales vs Profit')
plt.xlabel('Sales ($)')
plt.ylabel('Profit ($)')
plt.grid(True)
plt.tight_layout()
plt.show()
```



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In [10]: # Histogram: Distribution of Sales
plt.figure(figsize=(8, 5))
plt.hist(df['Sales'], bins=5, color='purple', edgecolor='black')
plt.title('Sales Distribution')
plt.xlabel('Sales ($)')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



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In [11]: # Box Plot: Profit Distribution
plt.figure(figsize=(8, 5))
plt.boxplot(df['Profit'], vert=False, patch_artist=True, boxprops=dict(facecol
plt.title('Profit Distribution')
plt.xlabel('Profit ($)')
plt.tight_layout()
plt.show()
```

Profit Distribution

